

The VR Wonderland of Yue Opera

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Abstract

As the second largest opera genre in China, Yue Opera's digital inheritance faces the dual challenges of a lack of interest from young people and insufficient interaction with existing technologies. To this end, this study proposes and constructs a virtual reality gamification system for the cultural inheritance of Yue Opera, "The VR Wonderland of Yue Opera" (VWYO). The system integrates virtual reality, motion capture, intelligent voice, and large language models to provide users with an immersive learning experience from cultural cognition, knowledge interaction, to performance imitation through the four-act scene of "voice dialogue - Yue Opera game - body learning - gesture learning". The results show that VWYO has advantages in improving the immersion of Yue Opera and helping to master knowledge, and also provides a generalizable technical path and practical paradigm for the digital living inheritance of intangible cultural heritage.

CCS Concepts

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Human-centered computing**; • **Interaction design**; • **Applied computing**; • **Arts and humanities**;

Keywords

Embodied Interaction, Gamified Learning, Motion Capture, Large Language Model, Intangible Cultural Heritage

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1 Introduction

Chinese opera culture is extensive and profound, with a long history, and Yue Opera is an important intangible cultural heritage. Most of the existing digital means are limited to two-dimensional media presentation, making it difficult to achieve deep cultural interaction [Damala et al., 2019].

Recently, cultural institutions have embraced the use of digital technologies to develop tools and platforms for promoting traditional cultures [Innocente et al., 2023; Trunfio et al., 2021]. The discussion of opera digitalization has entered the stage of paying equal attention to immersive experience and generative design from static archiving. The conceptual framework of "digital twin opera stage" is proposed [Cairolì and Tagliabue, 2023].

While gamification in education is known to improve engagement [Hallifax et al., 2019; Metwally et al., 2020], applying gamified learning to cultural contexts can make traditional art forms more accessible and engaging [Varinlioglu et al., 2022; Martusciello et al., 2025]. Museums, accordingly, follow a gamification approach that can provide a collaborative and entertaining learning experience to young visitors [Nofal et al., 2020]. Varinlioglu et al. explore using serious board games to simulate historical spatial experiences, such as replicating past trade route networks [Varinlioglu et al., 2023]. MillSecret is a multi-player Serious Escape Game designed to engage players with local cultural heritage through puzzles centered on the historical significance of watermills [Tzima et al., 2021]. The "Lion Dance Crafting Workshop" exemplifies the translation of traditional craft knowledge into an interactive gaming experience [Chai, 2024].

Virtual reality (VR) has opened up innovative ways for learning about cultural heritage. Selmanović et al. confirmed that 360-degree videos presented through virtual reality can help enhance user engagement [Selmanović et al., 2020]. The Leap Motion controller is used to capture the user's gestures in real time, allowing the user to perform a puppet show in the same virtual scene [Way and Wei, 2023]. The VR interactive system integrates three methods:

virtual avatar, participatory performance, and game-based knowledge acquisition, so that the oral performance of the intangible cultural heritage "Hua'er" can be passed on [Liu et al., 2022a]. In the VR scene, participants are guided to tap the target and experience "Dance of Drums" in a multimodal way in a real space [Liu and Yan, 2022b].

Inspired by these technologies [Rakkolainen et al., 2021], this study proposes an innovative framework that integrates gamification education and virtual reality, taking Yue Opera "Liang Zhu" as the practice object, and first constructing a gamified Yue Opera education system, which is called "The VR Wonderland of Yue Opera" (VWYO). The system is divided into four parts, namely voice dialogue, Yue Opera games, body learning, and gesture learning. The purpose is to allow users to actively learn Yue Opera culture from multiple aspects, then invite 30 participants to participate in the experiment, and finally analyze the results from the aspects of immersion, knowledge mastery, and cultural identity.

2 System Design and Development

"The VR Wonderland of Yue Opera" (VWYO) is a learning system designed for Yue Opera lovers, with the core concept of "gamified learning" and "intelligent interaction", through four acts of coherent scene narrative, including voice dialogue, Yue Opera games, body learning, and gesture learning. It integrates virtual reality (VR), motion capture, intelligent voice, large language models, and other cutting-edge technologies to build cultural cognition.

2.1 VR Scene

In scene 1, Users press the controller to record and ask questions to the virtual person Zhu Yingtai, forming a basic understanding of the historical background and cultural knowledge of the Yue Opera "Liang Shanbo and Zhu Yingtai".

In scene 2, the user finds the scroll according to the voice prompt. The puzzle appears after opening the scroll, and the user puzzles over the plum blossom patterns, deepening the understanding of the pattern structure and layout through interactive experience. After completing the task, high-fidelity 3D models of opera shoes appear in the scene, requiring users to grab the correct options and then follow the voice prompts to perform the next step. When the shoes are successfully identified, the user first hears the crisp magpie call, and then looks up to see the magpie flying on the plum blossom tree to guide the user. After arriving at the designated place, which is the city gate with the words "Hangzhou City" written on it, the door is closed, and the user must correctly perform all tasks of pace matching to let the door open. The puzzle game is shown in Figure 1a. The game of shoes and tasks of pace matching are shown in Figure 1b and Figure 1c.

During the user's journey, when encountering an unfolding Yue Opera fan, the user needs to select the corresponding objects in turn according to the order of the words in which the librettos are vacant, and complete the librettos. The corresponding librettos of the fan are the famous singing section "The Eighteen-Mile Farewell" by Liang Zhu of Yue Opera, and the objects mentioned in the singing section are all displayed in the scene. In the game, the user first starts from the study, then to the plum blossom tree, then to Hangzhou City, and finally to the pavilion, which is consistent with

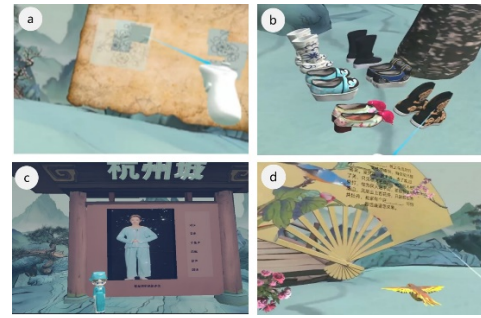


Figure 1: Yue Opera Game, a) Pattern puzzle, b) Shoe analysis, c) Pace matching, d) Fill in the libretto.

the path mentioned in the librettos. It is of great help to the user to understand the librettos. The game of libretto is shown in Figure 1d.

In scene 3, users watch the standard animation performance of the virtual avatar in the virtual environment, and learn the movements by watching the animation. They can choose to play one of the characters with the controller, and the user's key points are extracted through Mediapipe, interpolated, and smoothed, so that users can intuitively observe their body posture and pace during the performance process, and compare with the standard movements to correct their performance.

In scene 4, when the Yue Opera performance is completed, the user needs to further learn the relevant details of the Yue Opera gesture, and display 7 Yue Opera gesture pictures in front of the user for the user to imitate, these 7 custom gestures are common gestures in Yue Opera performances, named: Single Finger, Buddhist Fist, Tiger's Mouth Palm, Praise Finger, Raised Finger, Orchid Finger, Sword Fingers. During the game, the user's real-time gesture actions are collected to determine whether the corresponding parameters of the finger are within the threshold range, so as to judge whether the user has completed the corresponding gesture correctly and standardly. When the user completes the gesture learning, the butterfly guides the user to the designated place and gets the blind box reward.

2.2 Design of Intelligent Virtual Avatars

In designing the character images of "Liang Shanbo" and "Zhu Yingtai", we extensively drew inspiration from classic depictions of their appearances across various art forms. We also conducted in-depth research into the historical background and costume characteristics of the story's setting. We used Character Creator 4 for modeling. Character image of "Liang Shanbo" is shown in Figure 2a, and virtual avatar "Zhu Yingtai" is shown in Figure 2b. And the avatars are used in Unity 2022, which can be applied to Meta Quest 3.

To build virtual characters with intelligent interactive capabilities, we deeply integrated them with the large language model ChatGPT. This integration enables them not only to accurately understand and respond to participants' questions but also to support multi-turn continuous conversations.

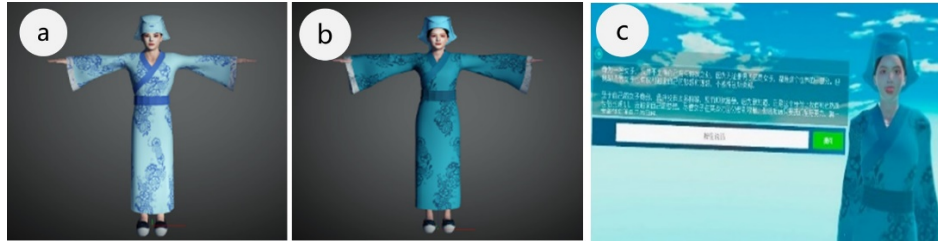


Figure 2: a) Virtual avatar “Liang Shanbo”, b) Virtual avatar “Zhu Yingtai”, c) Virtual avatar “Zhu Yingtai” in Unity.

Furthermore, to achieve synchronized “audio playback” and “lip-sync driving,” we employed Crazy Minnow Studio’s Salsa LipSync. This allows the virtual characters to move beyond mechanically playing sounds and instead exhibit lip movements highly synchronized with the speech content. SALSA LipSync controls lip shapes and movements through configured phoneme mouth shapes. The scene of performing intelligent conversations is shown in Figure 2c.

2.3 Production of Virtual Avatars’ Animation

Using inexpensive motion capture software (called DiMocap) to capture the movements of professional Yue Opera actors, it has the advantage of not being easily affected by the environment and not being obstructed, and then the captured action data is smoothed in Cascadeur to capture four acts of standard Yue Opera animation, including “Sworn Brotherhood at the Grass Bridge”, “The Eighteen-Mile Farewell”, “Reunion at” the Pavilion”, and “Metamorphosis into Butterflies”, and each scene is so detailed that it captures finger movements.

2.4 Data Processing and Interaction Processes

2.4.1 Body Learning. When the user wears the Meta Quest 3 helmet to participate in the Yue Opera performance, the eyes and part of the nose are obscured, resulting in the overall downward movement of the facial key points, and MediaPipe calculates the key points of the left eye, left ear, right eye, right ear, and nose by rotating the data instead of the position information in Unity. The design of “The VR Wonderland of Yue Opera” is shown in Figure 3. Our control group is called Baseline VR Yue Opera (BVYO). The Python side stores the data and sends the key point string. Unity obtains this information through UDP. We parse the 33 key points of MediaPipe in Unity, interpolate and calculate the additional bone points of Unity (such as spine, neck, hips, etc.), and then smoothly transition the key point data to avoid jitter, updating the position of the Unity bones and calculating based on quaternions. We eliminate the rotation effect of the initial T-Pose, and calculate the rotation value of the key points in the direction of the bone’s target. Finally, the virtual avatar can smoothly perform our actions.

2.4.2 Gesture Learning. Unity’s package called XR Hands provides precise position and rotation data for each bone, which is crucial for teaching that requires precise reproduction of the subtle angles and shapes of Yue Opera gestures. First, the standard hand shape performed by Yue Opera actors is collected. The error-prone points

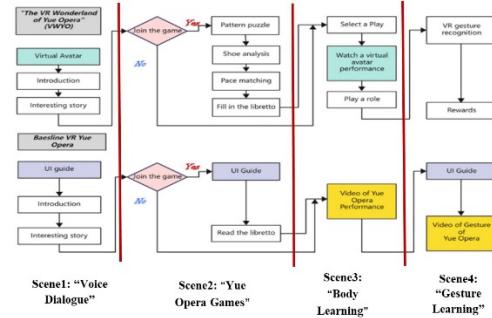


Figure 3: The design of “The VR Wonderland of Yue Opera”.

and attention points of each gesture are clarified, and the error-prone points are used as the key basis for identification. Then, the reasonable threshold range of the correct gesture is determined through multiple experiments. The user’s real-time gestures are read during gesture learning, and the system judges the gesture category and whether it is standard. We have created 7 new custom gestures, namely Single Finger, Buddhist Fist, Tiger’s Mouth Palm, Praise Finger, Raised Finger, Orchid Finger, and Sword Fingers.

3 User Study

3.1 Study Design

A total of 30 participants were invited to participate in the experiment. Before the start of the experiment, we provided all experimenters with a training session using the Meta Quest 3. After the training was completed, we divided the participants into two groups: the control group and the experimental group, each group consisting of 15 people. During the experiment, the participants in the experimental group used a system called VWYO, and the participants in the control group used the BVYO system. Both systems contain four different scenes: voice dialogue, Yue Opera games, body learning, and gesture learning. In VWYO, participants can engage in multimodal comprehensive learning through continuous dialogue with virtual humans, body key point recognition, and participants using the BVYO system can only understand Yue Opera culture through the user interface guide. The experience of “The VR Wonderland of Yue Opera” is shown in Figure 4.

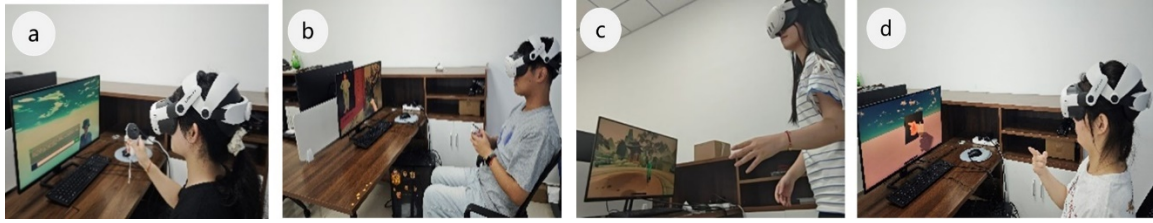


Figure 4: a) Experience of scene 1, b) Experience of scene 2, c) Experience of scene 3, d) Experience of scene 4.

Table 1: Post-Experiment Measurements and Descriptive Statistics

Dimension	Measurement Method	Description	Scale
Immersion	5-point Likert scale questionnaire	Measured the user's sense of presence and immersion in the virtual environment.	5-point Likert
Knowledge Test	6 multiple-choice questions	Assessed mastery of cultural knowledge of Yue Opera "Liang Zhu".	0-6 (Converted to 0-5)
Cultural Identity	5-point Likert scale questionnaire	Evaluated interest, willingness to inherit, and willingness to share Yue Opera art.	5-point Likert

3.2 Data Collection

Immediately after the experiment, participants completed a set of comprehensive questionnaires, in which the immersion part used a 5-point Likert scale to measure the user's sense of presence in the virtual environment; The knowledge test part contains 6 multiple-choice questions to examine the mastery of cultural knowledge such as the librettos, costumes and figures of the Yue Opera "Liang Zhu" (the original full score is 6 points, which has been converted to a 5-point system for ease of comparison); In the cultural identity part, users' interest, willingness to inherit and share Yue Opera art are evaluated through a 5-point Likert scale, which is shown in Table 1. In view of the small sample size, the mean (M) and standard deviation (SD) of each dimension were reported by descriptive analysis. Welch's t-test was used in this study.

3.3 Results

3.3.1 Language Dialogue. In this scenario, the VWYO system also shows strong effectiveness. By creating a realistic conversational context, the user's immersion ($t = 9.82, p < 0.001$) and knowledge mastery ($t = 7.28, p < 0.001$) are enhanced. Ultimately, this deep interactive experience also translates into a stronger cultural identity ($t = 9.70, p < 0.001$), indicating that virtual dialogue is a highly effective way to understand and internalize cultural connotations. Results of scene 1 are shown in Figure 5a.

3.3.2 Yue Opera Games. In this scene, the VWYO group showed an all-round improvement. The score in Immersion is significantly higher than the control group. Among them, the immersion score was 4.85 (SD = 0.32) vs. 3.95 (SD = 0.38). Concerning cultural identity, the VWYO group showed a mean improvement of 4.91 points (SD = 0.30), compared with 4.15 points (SD = 0.36) in the control group. This shows that VWYO's virtual narrative can be

very effective in stimulating emotional resonance and cultural belonging. At the same time, users also made statistically significant improvements in knowledge mastery ($t = 6.94, p < 0.001$), proving the success of this the scene of "Yue Opera Games" are shown in Figure 5b.

3.3.3 Body Learning. The results of this scene present an interesting contrast. The VWYO group is immersive. In terms of immersion and cultural identity score, the immersion score was 4.75 (SD = 0.33) vs. 3.85 (SD = 0.40). The cultural identity score was 4.80 (SD = 0.32) vs. 3.85 (SD = 0.38), which indicates that the virtual environment optimizes the learning experience. However, there was no significant difference between the two groups in knowledge mastery ($t = 1.44, p = 0.155$), suggesting that the added value brought by immersive experience for basic motor skills could not be directly translated into immediate operational knowledge advantages, but the user's score on cultural identity increased ($M = 4.80$), which was conducive to strengthening interest and beneficial to long-term learning in the future. Results of the scene of "Body Learning" are shown in Figure 5c.

3.3.4 Gesture Learning. The gesture learning scenario highlights the value of the VWYO system. It achieved statistical significance advantages in immersion ($t = 9.85, p < 0.001$), knowledge mastery ($t = 6.43, p < 0.001$), and cultural identity ($t = 10.40, p < 0.001$). This suggests that the system can provide more effective guidance and understanding in imparting subtle gestures rich in cultural meaning than traditional demonstrations. Results of the scene of "Gesture Learning" are shown in Figure 5d.

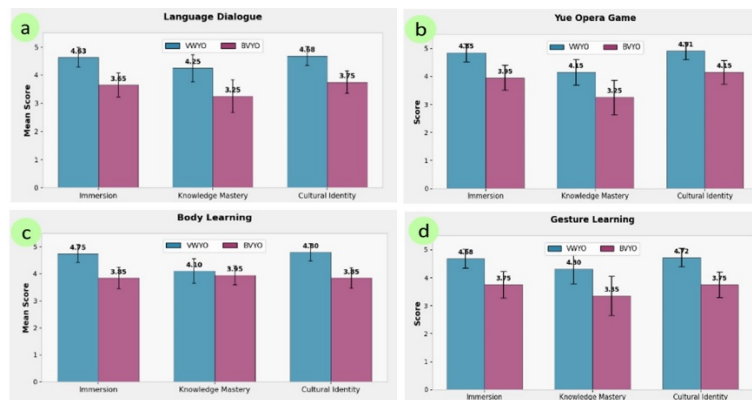


Figure 5: a) Results of Scene 1: “Language Dialogue”, b) Results of Scene 2: “Yue Opera Games”, c) Results of Scene 3: “Body Learning”, d) Results of Scene 4: “Gesture Learning”.

4 Conclusion and Future Work

The experimental results show that VWYO outperforms the traditional learning method (BVYO) in the three dimensions of immersion, knowledge mastery, and cultural identity, which verifies the effectiveness and innovation potential of virtual reality technology and gamification mechanisms in the inheritance of intangible cultural heritage. In the future, the experimental scope will be expanded, and a diverse range of users will be recruited through long-term experiments. We will improve the level of technical intelligence and integrate multimodal dialogue technology to strengthen the emotional interaction ability. Also, we will migrate the system framework to other ICH projects. Future research will conduct a categorized analysis of representative ICH types to clarify their core cultural expression forms and differences in interactive needs, and then refine targeted technical adaptation pathways, such as optimizing motion capture algorithms and developing haptic feedback modules. Through these measures, we can promote the development of cultural content.

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